MD. ZAHIDUL HAQUE (ALVI)

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Ph.D. Researcher — AI for Software Engineering

Summary

Ph.D. researcher at William & Mary working at the intersection of large language models and automated software engineering. Published in ACM Transactions on Software Engineering and Methodology (TOSEM). Experienced in parameter-efficient fine-tuning (LoRA/QLoRA), multi-task learning, and systematic evaluation of large code models. Built end-to-end training and evaluation pipelines for code generation, translation, and summarization tasks.

Education

- **William & Mary** — Williamsburg, VA, USA
  ***Ph.D. Student in Computer Science**; GPA: 3.92/4.00* — *Aug 2023 - Present*
  - **Research Group**: AURA Lab (Advisor: Prof. Antonio Mastropaolo).
  - **Research Focus**: LLM optimization and multi-task learning at the intersection of AI and Software Engineering, specializing in **Parameter-Efficient Fine-Tuning (PEFT)** for **Large Code Models**.
  - **Relevant Coursework**: Software Maintenance & Evolution, Deep Learning, AI for Software Engineering, Autonomous Driving & Connected Mobility, Deep Transfer Learning, Data Mining, Human-Centered Computing.
- **Sylhet Engineering College (SEC)** — Sylhet, Bangladesh
  Bachelor of Science in Computer Science & Engineering; GPA: 3.46/4.00 — *Mar 2016 - Mar 2021*

Publications

- **Parameter-Efficient Multi-Task Fine-Tuning in Code-Related Tasks**
  ***Md. Zahidul Haque**, Saima Afrin, Antonio Mastropaolo*
  ACM Transactions on Software Engineering and Methodology (TOSEM), Aug 2026.
  [Preprint] **[Code]**
  - Investigated **Multi-task QLoRA** fine-tuning across code generation, translation, and summarization. First study to jointly analyze functional correctness and non-functional code quality via static analysis and **LLM-as-Judge** in parameter-efficient multi-task settings.

- **A Systematic Literature Review of Parameter-Efficient Fine-Tuning for Large Code Models**
  Saima Afrin, **Md. Zahidul Haque***, Antonio Mastropaolo*
  ACM Transactions on Software Engineering and Methodology (TOSEM), Oct 2025.
  [Paper] **[Code]** (*Equal contribution)
  - Analyzed 28 peer-reviewed studies on **PEFT** techniques (LoRA, Adapters, Prefix Tuning) applied to large code models (CodeLlama, StarCoder, DeepSeek-Coder). Produced a comprehensive taxonomy of adaptation strategies across generative and non-generative SE tasks.

Technical Skills

- **Programming**: Python, C++, C, SQL, Bash
- **LLM & Model Adaptation**: Hugging Face Transformers, PEFT (LoRA/QLoRA/Adapters), Multi-task Learning, RAG, Quantized Fine-Tuning (4/8-bit), vLLM
- **Model Training & Systems**: PyTorch, TensorFlow, Scikit-learn, Multi-GPU Training, HPC/Slurm, Docker, Linux
- **Evaluation & Analysis**: pass@k, BLEU, ROUGE, CodeBLEU, Static Analysis, Statistical Testing (Wilcoxon, Cliff's Delta)

Professional Experience

- **Halliburton – Landmark** — Houston, TX
  Data Scientist Intern — *May 2026 - Aug 2026*
  - Built and deployed an agentic LLM system that answers natural-language questions as SQL over a large enterprise engineering database, delivered as a live chat interface for internal users.
  - Designed its retrieval, schema-linking, and evaluation components, substantially improving end-to-end query accuracy over prior internal prototypes.
- **William & Mary – AURA Lab** — Williamsburg, VA
  Graduate Research Assistant — *Aug 2023 - Present*

- Designed and implemented end-to-end fine-tuning and evaluation pipelines for large code models across code generation, translation, and summarization tasks. This work resulted in one publication in ACM Transactions on Software Engineering and Methodology (TOSEM) and one additional submission currently under review at TOSEM.
- Currently investigating security-oriented aspects of large language models for code, including secure code generation and fine-tuning strategies for non-generative code analysis tasks.

- **Graaho** Reston, VA (Remote)
  Software Engineer (Machine Learning) *Mar 2022 - Sep 2022*
  - Designed and implemented an end-to-end machine learning pipeline for credit risk assessment, including data cleaning, feature engineering on large-scale financial datasets, baseline modeling (Logistic Regression, Random Forest, XGBoost via Scikit-learn), and automated testing with Pytest. Developed reproducible, version-controlled evaluation workflows to support financial compliance.
- **Mayalogy** Dhaka, Bangladesh
  Research Engineer (Machine Learning) *Feb 2021 – Feb 2022*
  - Developed *MayaNERC*, a biomedical named entity recognition pipeline to extract 17 entity types from patient labels using transformer-based models, validated across 111 diseases spanning five clinical categories.
  - Built three production NLP systems–Auto Tag, Group Referrer, and a spam filter for Romanized Bangla and Urdu text–achieving Macro-F1 scores in the high-80% range, and deployed for real-time content moderation.
  - Led a cross-functional team of 12+ to curate high-quality deep learning datasets; designed annotation guidelines achieving inter-annotator agreement (Cohen's Kappa) of approximately 0.82. Collaborated closely with medical experts to support AI-assisted healthcare applications.

Research Projects

- **B-NER: A Novel Bangla Named Entity Recognition Dataset (IEEE Access, 2023)**
  First Author · *[Paper]* *[Code]* *Apr 2023*
  - Created the largest Bangla named entity recognition dataset, consisting of 22,144 manually annotated sentences with 9,895 unique words across eight entity types sourced from newspapers and Wikipedia.
  - Benchmarked BiLSTM with fastText embeddings, sentence transformers, and fine-tuned IndicBERT, achieving 86% Macro-F1. Conducted cross-dataset transfer evaluation demonstrating superior generalization over existing Bangla NER datasets.

Teaching Experience

- **William & Mary** Williamsburg, VA
  Graduate Teaching Assistant *Fall 2023 - Present*
  - **Courses**: Generative Artificial Intelligence for Software Development (CSCI 455), Prompt Engineering (CSCI 100), Principles of Programming Languages (CSCI 312), Intro to Programming (CSCI 141).
  - **Contributions**: Built lab modules and assignments on RAG systems, n-gram modeling, repository mining for dataset construction, and pretrained model usage. Conducted lab sessions and office hours for 30+ students across Python and C++ curriculum.

Honors & Certifications

- **Robi Datathon 2.0** – Pre-Assessment on Tabular Data Analysis (2022) | **DL Sprint, BUET CSE Fest** – Bengali ASR Competition (2022) | **Cisco CCNA** – Certified Network Associate (2020–2023)